

Program: Diploma in Civil and Rural Engineering	
Course Code:	Course Title: Rural Infrastructure and Planning Lab
Semester: 3	Credits: 1.5
Course Category: Program Core	
Periods per week: 3 (L:0 T:0P:3)	Periods per semester: 45

Course Objectives:

- To plan the strategies for successful implementation of infrastructure projects in rural area.

Course Pre requisites:

Topic	Course code	Course name	Semester
Basic computer knowledge		Introduction to ITsystems	S1
Knowledge of Engineering Graphics		Engineering Graphics	S1

Course Outcomes:

On completion of the course, the student will be able to:

CO n	Description	Duration (Hours)	Cognitive Level
CO1	To prepare a report of the basic infrastructure facility built under watershed management program in a nearby village	9	Applying
CO2	To understand the basic infrastructure facility built under various Govt schemes in a rural area by field visit	6	Applying
CO3	Develop plan, elevation and section of various type of rural buildings	15	Applying
CO4	Develop rainwater harvesting drawing, septic tank drawing and service plan of the building	9	Applying
	LabTests	6	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1				3			
CO2				3			
CO3				3			
CO4				3			

3-Strongly mapped, 2-Moderately mapped,1-Weakly mapped

Course Outline:

On completion of the course, the student will be able to:

Module Outcomes	Description	Duration (Hours)	CognitiveLevel
CO1	To prepare a report of the basic infrastructure facility built under watershed management program in a nearby village		
M1.01	Prepare a report of weir and underground bandhara of a nearby village by field visit	6	Applying
M1.02	Prepare a report of contour bunding in the nearby village by field visit	3	Applying
CO2	To understand the basic infrastructure facility built under various Govt schemes in a rural area by field visit		
M2.01	Prepare the report, including your comments on construction and working of Gobar gas/biogas plant, if any after undertaking the visit to the biogas plant in the nearby rural area.	3	Applying
M2.02	Prepare the report, including your comments on construction and working of solar power panel if any after undertaking the visit to the biogas plant in the nearby rural area.	3	Applying
CO3	Develop plan, elevation and section of various type of buildings in rural area		
M3.01	Develop plan, elevation and sectional views of single storied and two storied rural house	6	Applying
M3.02	Draw the plan, elevation and section of a rural hospital building	3	Applying

M3.03	Draw the plan, elevation and section of a school building in rural area	3	Applying
M3.04	Draw the plan, elevation and section of a public office building in the rural area	3	Applying
	Lab test I	3	
CO4	Develop rain water harvesting drawing, septic tank drawing and service plan of the building		
M4.01	Develop detailed drawing of rainwater harvesting	3	Applying
M4.02	Develop detailed drawings of septic tank	3	Applying
M4.03	Develop service plan of the building	3	Applying
	LabTest II	3	

Text/Reference:

T/R	BookTitle/Author
T1	Auto CAD Essentials,Autodesk official Press,JohnWiley&Sons,USA
R2	Computer Aided Design Laboratory by M.N.SeshaPraksh& Dr.G.S.Serveshe –Laxmi Publications.
R3	Engineering Graphics byP. J.Sha– S. Chand &Co.
R4	Introduction to AutoCAD 2018 for Civil Engineering Applications by Nighat Yasmin,SDC Publications

Online Resources:

Sl.No	WebsiteLink
1	https://www.autodesk.in/campaigns/autocad-tutorials
2	https://thesourcecad.com/autocad-tutorials/
3	https://all3dp.com/1/autocad-tutorial-for-beginners/